

IVIVC App User Guide

Contents

1. Introduction	2
2. What the app does	2
3. Inputs	2
4. Outputs	3
5. Input datasets	3
6. Analysis approaches and models	3
6.1 Approaches	3
6.2 Models	4
7. Preprocessing options	6
7.1 Normalization	6
7.2 Scaling	6
7.3 Interpolation	6
8. Performance metrics	7
9. Cross-validation options	8
9.1 Choosing a cross-validation scheme	9
9.2 Cross-validation parameters	9
10. Grid-search initialization	10
11. Relative model evidence	10
12. Uncertainty and error bars	11
12.1 Approach 1: Direct Correlation	11
12.2 Approach 2: Independent Model Fitting	12
12.3 Approach 3: Direct Mapping	12
13. Command-line interface	12
14. Extending the app	13
Appendix	14
A. How to Interpret Model Performance	14
A.1 Confirm that models are comparable before interpreting metrics	14
A.2 Metric ranges, direction, and interpretation	14
A.3 Interpreting goodness of fit	15
A.4 Interpreting prediction error	15
A.5 Interpreting relative model evidence	16
A.6 Interpreting cross-validation performance	16

A.7 Interpreting residual diagnostics	17
A.8 Interpreting parameter uncertainty	17
A.9 Considering model complexity	18
A.10 Scientific plausibility and intended use	18
A.11 Suggested model-selection workflow	19
A.12 Questions to Guide Model Selection	19
A.12.1 Does the model describe the observed data well?	19
A.12.2 Does the model appear to generalize beyond the fitted data?	19
A.12.3 Are the model parameters interpretable and stable?	19
A.12.4 Is the model unnecessarily complex?	20
A.12.5 Is the model scientifically plausible for the degradation process?	20
A.12.6 Is the model appropriate for the intended use?	20
A.13 Example interpretation statements	20
A.14 Limitations	21

References	22
-------------------	-----------

1. Introduction

Establishing an in vitro-in vivo correlation (IVIVC) can support the evaluation of medical devices that include absorbable polymers by relating in vivo changes in material properties over time to corresponding in vitro test results, such as degradation profiles. IVIVCs may help evaluate manufacturing or material changes, support predictions of in vivo performance when modifying an existing device design, and reduce reliance on animal studies during preclinical development. This tool is intended to facilitate IVIVC development. The user supplies paired in vitro and in vivo fitting data, and may optionally supply an in vitro prediction dataset. The app applies selected preprocessing steps, fits and compares multiple correlation approaches and model families, evaluates selected performance metrics and cross-validation results, generates fitting and prediction plots, and produces report-ready summaries that can be reviewed and incorporated into external documentation.

For general background on credibility assessment and reporting of computational modeling studies in medical device submissions, see FDA guidance on computational modeling credibility and reporting [1, 2].

2. What the app does

- Applies selected normalization, scaling, and interpolation options to prepare fitting and prediction data.
- Fits user-selected IVIVC approaches and model families to paired in vitro and in vivo data.
- Calculates selected goodness-of-fit, model-comparison, and cross-validation metrics.
- Generates fitting, comparison, and prediction plots with uncertainty displays where supported.
- Creates summaries for user-selected final approaches or models.

3. Inputs

The IVIVC app requires paired in vitro and in vivo degradation data, along with user-selected analysis settings that define how the data are processed, modeled, and summarized. At a high level, the inputs include:

- a spreadsheet containing in vitro and in vivo time-course degradation data for fitting correlations;
- optionally, a spreadsheet containing in vitro data from which to generate predictions using fitted correlations;
- selection of one or more IVIVC modeling approaches;
- selection of candidate models to evaluate, where applicable;
- optionally, preprocessing options, such as normalization, interpolation, or scaling choices;

- selected performance metrics to display in the results and reports; by default, the app selects R^2 , RMSE, and AIC_c as a simple starting set that has been useful in internal testing with real datasets but may not be optimal for every analysis;
- optionally, cross-validation settings for applicable parametric models; and
- grid-search settings for model initialization.

The sections below provide additional detail on input file structure, required data columns, preprocessing options, model-selection inputs, and optional analysis settings.

4. Outputs

The IVIVC app generates interactive results and downloadable reports that summarize fitted model behavior, model-comparison metrics, diagnostic information, and user-selected analysis settings. Output availability depends on the selected approaches, whether fitting succeeds, and whether a prediction dataset was supplied.

Output	Description
Preprocessing outputs	Processed data tables and plots showing the data used for fitting and prediction.
Model fitting outputs	Parameter estimates reported as estimate $\pm$ approximate 95% confidence-interval half-width where available, fitted plots, residual-vs-time plots, residual Q-Q plots, and selected performance metrics.
Model comparison outputs	Tables comparing selected approaches, models, metrics, and cross-validation summaries.
Prediction outputs	Prediction plots and downloadable tables when a prediction dataset is supplied and the selected method can be applied.
Selected model & summary of outputs	User-selected report summaries and optional Word report export for incorporation into external documentation.

5. Input datasets

The app uses separate fitting and prediction datasets. The fitting dataset is required because it provides the paired in vitro and in vivo data used to establish the IVIVC relationship; the prediction dataset is optional and is used only when applying a fitted relationship to new in vitro observations.

Input	Description
Fitting dataset	Required four-column dataset used to establish the in vitro/in vivo relationship. Columns are interpreted as in vitro time, in vitro value, in vivo time, and in vivo value.
Prediction dataset	Optional two-column in vitro dataset used after fitting to generate in vivo predictions. Columns are interpreted as in vitro time and in vitro value.

For Excel workbooks, select the sheet containing the relevant data. For CSV files, the first columns are read directly in the order described above.

6. Analysis approaches and models

In this app, an analysis approach is the overarching method used to establish or evaluate an in vitro/in vivo correlation. A model is the mathematical form used within an approach when a fitted equation is needed to describe the relationship.

6.1 Approaches

Approaches differ in how the in vitro and in vivo time courses are related to one another.

Approach	Description	Available models
Approach 1: Direct Correlation	Approach 1 directly relates matched in vitro and in vivo time points after preprocessing. The fitted model is then used as a time-scaling function that maps an in vitro time to a predicted in vivo time. This approach can be useful when the two datasets have matching time points, or when interpolation can create matched pairs, and the main question is how in vitro time corresponds to in vivo time.	Double Exponential, Exponential, Linear, Linear (zero intercept), Power
Approach 2: Independent Model Fitting	Approach 2 fits the same model family separately to the in vitro and in vivo datasets. With fitting data alone, the report summarizes the two fitted models and their fitted-curve uncertainty. When prediction data are supplied, predictions can be generated by comparing the fitted in vivo and in vitro responses or by comparing fitted characteristic time constants when those values are available. This approach can be useful when the two datasets were measured on different time schedules, when each dataset is expected to follow the same type of degradation curve, or when a specific model form is scientifically meaningful for both datasets.	Double Exponential, Exponential, Mass Loss (Gompertz), Mass Loss (One-Phase), Mass Loss (Surface Erosion), Mass Loss (Two-Phase), Power, Stretched Exponential
Approach 3: Direct Mapping	Approach 3 uses direct interpolation-based mappings between datasets rather than fitting a parametric model. It can describe value-ratio behavior over time and time-ratio behavior over value ranges. When prediction data are supplied, those mappings can be used to scale prediction values or prediction times. This approach can be useful for exploratory comparisons, for datasets where a simple equation does not describe the observed pattern well, or when the user wants a direct data-based mapping rather than a fitted model.	No parametric model selection

6.2 Models

Candidate models are the mathematical equations available for fitting within the applicable approaches. Different model forms represent different assumptions about the shape of the degradation relationship, such as linear behavior, power-law behavior, or sigmoidal transition behavior.

Model	Used in	Functional form	Notes
Double Exponential	Approach 1, Approach 2	$y = ae^{bx} + ce^{dx}$	Sum of two exponential terms. It can represent profiles with two apparent rates, but the added flexibility may require more data to estimate stably.

Model	Used in	Functional form	Notes
Exponential	Approach 1, Approach 2	$y = ae^{bx}$	Two-parameter exponential relationship. It can describe monotonic change that accelerates or decelerates exponentially over the fitted range.
Linear	Approach 1	$y = a + bx$	Straight-line empirical relationship with an intercept and slope. It is appropriate when the in vitro/in vivo relationship is approximately linear over the fitted range.
Linear (zero intercept)	Approach 1	$y = ax$	Straight-line empirical relationship constrained to pass through zero. It may be useful when a zero in vitro value should correspond to a zero in vivo value under the selected preprocessing and response scale.
Mass Loss (Gompertz)	Approach 2	$y = y_0 - A \exp\{-\exp[(er_{max}/A)(t_{lag} - x) + 1]\}$	Mass-remaining model based on the modified Gompertz curve. It is intended for asymmetric sigmoidal mass-loss profiles with an apparent lag or plateau, a maximum loss-rate region, and a terminal loss extent.
Mass Loss (One-Phase)	Approach 2	$y = y_0 - A/(1 + e^{-(x-t_{50})/s})$	Mass-remaining model with a delayed sigmoidal loss phase. It is intended for mass-loss profiles that begin constant and show a later drop.
Mass Loss (Surface Erosion)	Approach 2	$y = y_0 \max(1 - kx, 0)^n$	Mass-remaining model for surface-controlled erosion. The fitted exponent n allows the profile shape to adapt to slab-like, cylindrical, spherical, or empirical geometry-like erosion behavior.
Mass Loss (Two-Phase)	Approach 2	$y = y_0 - B(1 - e^{-k_b x}) - D/(1 + e^{-(x-t_{50})/s})$	Mass-remaining model with an optional early burst-loss phase followed by a delayed sigmoidal loss phase. It is intended for mass-loss profiles that may show an initial drop, an intermediate plateau, and a later drop.

Model	Used in	Functional form	Notes
Power	Approach 1, Approach 2	$y = ax^b$	Power-law empirical relationship. It can describe nonlinear monotonic relationships where the response changes as a fitted power of time or the mapped predictor.
Stretched Exponential	Approach 2	$y = ae^{bx^c}$	Exponential relationship with a fitted stretching exponent. It can describe non-single-rate behavior, but the additional shape parameter can increase sensitivity when data are sparse.

Mass loss model notes

The surface-erosion mass-loss form is based on geometry-controlled surface erosion concepts such as the erodible slab, cylinder, and sphere model [3]. The one-phase mass-loss model uses a logistic/sigmoidal transition to represent a delayed loss phase [4]. The two-phase mass-loss model combines an early burst-like loss term with a delayed sigmoidal loss phase. Burst and delayed/multiphase behavior in degradable polymer systems is discussed in the polymer erosion and PLGA release literature [5, 6, 7]. The Gompertz mass-loss model uses a modified Gompertz-style sigmoidal form with lag and maximum-rate parameters [8].

7. Preprocessing options

Preprocessing prepares the uploaded data for comparison, fitting, and prediction. Normalization expresses values relative to a reference value so datasets can be compared on a common relative scale. Scaling changes the units or magnitude of a variable without changing the underlying pattern. Interpolation estimates values at time points that were not directly observed, which may be useful when an approach requires matching in vitro and in vivo time points (i.e., Approach 1, Approach 3).

7.1 Normalization

Option	Internal key	Description
Min-max normalization	min_max	Converts response values to a 0-1 scale using the observed minimum and maximum within each dataset.
Initial-value normalization	initial_value	Expresses each response value relative to the first response value in the corresponding dataset.

7.2 Scaling

Option	Internal key	Description
Log10 time scaling	log_x	Uses a base-10 log scale for time values before fitting or interpolation.
Log10 response scaling	log_y	Uses a base-10 log scale for response values before fitting or interpolation.

7.3 Interpolation

Option	Internal key	Description
No interpolation	none	Uses only the original measured points. Values between measurements are not estimated, so unmatched positions are marked as missing.
Shared-time interpolation	default	Uses linear interpolation to estimate both datasets at a shared set of experimental time points. Values outside the measured range are marked as unavailable.
Densified interpolation	alternative	Adds extra evenly spaced points between adjacent observations before interpolation. This can smooth displays and make fitting more stable, but the added points are estimated rather than directly measured and can affect metric values.

8. Performance metrics

Performance metrics are summary measures that help describe how closely a fitted model matches the observed data, how large the prediction errors are, and, for some metrics, how well a model balances fit quality with model complexity. The following performance metrics are included in the app.

Metric	Internal key	Preferred direction	Description
R^2	r_squared	Higher is better	Coefficient of determination. Higher values indicate that the fitted curve follows more of the observed response pattern.
Adjusted R^2	adjusted_r_squared	Higher is better	R^2 adjusted for the number of fitted parameters. It can decrease when extra parameters do not improve the fit enough to justify the added complexity.
MSE	mse	Lower is better	Average squared difference between observed and fitted values. Lower values indicate smaller model errors, but the squared units can be harder to interpret directly.
RMSE	rmse	Lower is better	Typical model error in the same response units as the fitted data. Lower values indicate predictions closer to the observations.
MAE	mae	Lower is better	Average absolute model error in the same response units as the fitted data. Lower values indicate predictions closer to the observations.
NRMSE (mean)	mnmse	Lower is better	RMSE divided by the response mean. Lower values indicate smaller error relative to the typical response magnitude.
NRMSE (std. dev.)	nrmse	Lower is better	RMSE divided by the response standard deviation. Lower values indicate smaller error relative to the amount of variation in the response.

Metric	Internal key	Preferred direction	Description
AIC	aic	Lower is better	Akaike Information Criterion. AIC compares fitted models by considering both fit quality and the number of fitted parameters. Among comparable fitted models, a lower AIC value generally indicates better empirical support for a model that fits the data without unnecessary complexity. Absolute AIC values, including negative values, should not be interpreted as standalone quality grades.
AIC _c	aicc	Lower is better	Corrected Akaike Information Criterion. AIC _c is AIC with an added small-sample correction and is often preferred when the number of fitted observations is limited relative to the number of fitted parameters. See the user guide for additional information. Among comparable fitted models, a lower AIC _c value generally indicates better empirical support for a model that balances fit quality and complexity; AIC _c is undefined when there are too few observations relative to the number of parameters.
BIC	bic	Lower is better	Bayesian Information Criterion. BIC compares fitted models by considering both fit quality and the number of fitted parameters, usually applying a stronger penalty for extra parameters than AIC. Among comparable fitted models, a lower BIC value generally indicates better empirical support for a model that fits the data without unnecessary complexity. Absolute BIC values, including negative values, should not be interpreted as standalone quality grades.

For additional background on common error metrics, scale-dependent accuracy measures, and adjusted R^2 , see references [9, 10, 11].

For AIC_c, a commonly cited rule of thumb is to prefer AIC_c over AIC when the ratio of fitted observations to estimated parameters is small, such as $n/k < 40$, where n is the number of fitted observations and k is the number of model parameters [12].

By default, the app selects R^2 , RMSE, and AIC_c. This compact default set has been simple and useful in internal testing with real datasets, but it is not necessarily the best set for every dataset or intended use. Users may select additional metrics when they provide useful context for a specific analysis. The default set gives three complementary views of model performance. R^2 is widely recognized and describes how well the fitted curve follows the observed response pattern. RMSE reports typical error in the original response units, making it easier to judge whether the error is practically large or small. AIC_c supports comparison among fitted candidate models by balancing fit quality with a penalty for unnecessary model complexity, with a correction that is useful for smaller datasets.

9. Cross-validation options

When developing a model, we want to know how well it will perform on new data it has not seen before. Cross-validation (CV) estimates how a fitted model predicts observations that were not used to fit that model. To do this, the app repeatedly sets aside part of the data as test data, fits the model using the remaining training data, and then compares the model predictions with the test data. In this app, the cross-validation result is best interpreted as a sensitivity diagnostic based on prediction error under the selected splitting pattern. It is not a confidence interval or a parameter uncertainty estimate. When final-fit error from the model fitted to the entire dataset and cross-validation error are similar, this provides some reassurance that the fitted relationship is not overly dependent on the particular observations used for fitting. When cross-validation error is much worse, the model may be more sensitive to the available data, the selected splitting scheme, or the number and placement of time points.

For time-course data, the splitting pattern matters. Setting aside one time point, a short contiguous time window, or a random subset represents a different artificial prediction task. Nearby observations may contain related information, and early, middle, and late regions of an absorbable-polymer degradation profile may reflect different parts of the degradation process. A split that leaves training observations immediately before and after an omitted observation may mainly test interpolation within the observed curve, while a contiguous-block split asks whether the model can bridge a missing region.

There is no universally best cross-validation scheme for every IVIVC dataset. The most useful choice depends on the practical question being asked and on whether enough observations remain for stable fitting. When results change substantially across schemes, treat that as a signal to review fitted plots, residuals, parameter uncertainty, model complexity, and scientific plausibility rather than as an automatic reason to accept or reject a model.

For background on cross-validation procedures and their use in model assessment and selection, including issues that arise with temporally ordered data, see references [13, 14, 15, 16].

9.1 Choosing a cross-validation scheme

The table below summarizes the schemes available in the app. The “CV error describes” column describes the type of prediction error or sensitivity each scheme most directly evaluates.

Scheme	CV error describes	Useful when	Main caution
Leave-one-timepoint-out	Single-timepoint prediction error; sensitivity to removing one observation at a time.	Default starting point for small time-course datasets because each repeated fit retains nearly all observations.	May be a gentle test when neighboring time points strongly constrain the omitted point.
Leave-contiguous-block-out	Local time-window prediction error; sensitivity to omitting a short adjacent region of the profile.	Main time-aware sensitivity check when adjacent observations may be related or when a missing time region is a concern.	Larger blocks are more demanding but may leave too few observations.
K-fold	Ordered fold prediction error; sensitivity to setting aside larger ordered portions of the dataset.	Familiar fold-based diagnostic when there are enough observations to form meaningful folds. Smaller k values create larger test regions.	Training data for a test fold may include time points on either side of it. Results can depend on fold boundaries.
Shuffle split	Random-subset prediction error; sensitivity to repeated random training/test splits.	Exploratory robustness check when temporal ordering is not central or when conclusions should be compared across random subsets.	Least time-aware option; can give optimistic scores when nearby observations are strongly related.

Scheme	CV error describes	Useful when	Main caution
None	No cross-validation error is calculated; final fitted-model summaries are reported without a repeated prediction check.	Appropriate when there are too few observations for meaningful CV, repeated fits are unstable, or only fitted-model behavior is needed.	Removes one robustness diagnostic; rely more heavily on fitted profiles, residuals, final metrics, parameter diagnostics, and scientific plausibility.

9.2 Cross-validation parameters

Parameter	Internal key	Description
Cross-validation scheme	cv_scheme	Selects the splitting method used to estimate cross-validation performance for fitted parametric models. Leave-one-timepoint-out is the default.
CV splits/folds/block size	cv_n_splits	For Shuffle split, the number of repeated random train/test splits. For K-fold, the number of folds. For Leave-contiguous-block-out, the number of adjacent time points set aside per split. The setup page does not enforce a maximum block size because the uploaded data are not read until Analyze is selected; larger blocks may leave too few observations for stable fitting. This setting is not used for Leave-one-timepoint-out or None.
Shuffle split test size	cv_test_size	The fraction of observations used as test data in each Shuffle split repeat. Smaller values leave more observations for fitting; larger values make the test more demanding but leave fewer observations for training.
CV random seed	cv_random_state	Seed used to make randomized cross-validation splits reproducible. This applies to Shuffle split.

10. Grid-search initialization

The app uses grid-search initialization to help start fitting from reasonable parameter values for each model. It tries many candidate starting-value sets within the configured bounds and uses the best-performing candidate to start the final curve fit.

This step reduces the need for manual parameter tuning and can make fitting more stable, but it is not itself the final fitting algorithm and does not guarantee that the final fitted parameters are the global optimum.

The default grid-search settings can usually be left unchanged. Consider adjusting these settings when fits are unstable, runtime is excessive, or a reproducible analysis configuration is needed.

Parameter	Internal key	Description
Grid-search random starts	grid_search_num_points	Number of candidate starting parameter sets evaluated before nonlinear least-squares fitting. Higher values may improve robustness but increase runtime.
Grid-search CPU cores	grid_search_num_cores	Number of CPU cores used for the starting-point search. Only available via command-line interface.
Grid-search parameter minimum	grid_search_param_min	Lower bound for randomly sampled candidate starting parameter values.

Parameter	Internal key	Description
Grid-search parameter maximum	grid_search_param_max	Upper bound for randomly sampled candidate starting parameter values.
Grid-search random seed	grid_search_random_state	Seed used to make randomized starting-point sampling reproducible. Changing the seed changes the sampled candidates while keeping the same bounds and number of starts.

11. Relative model evidence

Relative model evidence compares candidate parametric models fitted to the same dataset under the same preprocessing settings and IVIVC approach. In this guide, empirical support means support, within the displayed candidate set, that a model best balances fit quality and complexity. The evidence table is based only on the information-criterion metrics selected by the user: AIC, AIC_c, and/or BIC. These criteria are relative; their absolute values are not interpreted directly, and negative values are allowed.

Table definitions: n is the number of fitted observations; k is the number of fitted model parameters; Delta is the difference between a model's criterion value and the lowest value in the displayed candidate set; weight is the normalized relative support for that model under the criterion; and evidence ratio compares the highest weight in the set with the model's weight. Delta values near 0 indicate similar relative support; larger delta values indicate lower support relative to the leading candidate under that criterion.

Similar empirical support does not mean the models are scientifically equivalent. Final model selection should also consider prediction error, cross-validation behavior, residual patterns, parameter uncertainty, model simplicity, and biological or degradation-mechanism plausibility.

The app reports these summaries to support user review. It does not select, recommend, or validate a model.

For additional background on information criteria, empirical support, and evidence-weight interpretation, see references [17, 18, 12, 19, 20].

Support summary	Interpretation
Delta	Difference from the lowest value in the displayed candidate set. Values near 0 indicate similar relative support; larger values indicate lower support relative to the leading candidate under that criterion.
Weight	Normalized relative-support value across the candidate models in the displayed evidence table. Weights are relative, not absolute probabilities that a model is true.
Evidence ratio	Ratio of the highest evidence weight to the model's evidence weight. Larger ratios indicate less relative support compared with the highest-weighted model in the set.
Interpretation badge	Neutral rule-of-thumb label based on the displayed delta values. The label supports user review and is not an app recommendation.

12. Uncertainty and error bars

The following descriptions summarize how fitted-parameter uncertainty, fitting-plot bands, and prediction error bars are calculated or interpreted for each approach.

12.1 Approach 1: Direct Correlation

Fitting uncertainty

Parameter estimates: fitted parameters are reported as estimate $\pm$ the approximate 95% confidence-interval half-width. The half-width is calculated from the nonlinear least-squares parameter covariance matrix as $t(0.975, \text{residual})$.

degrees of freedom) multiplied by the parameter standard error. Correlations among fitted parameters are retained when uncertainty-aware parameter values are created.

Fitting plot bands: shaded 95% prediction bands are generated by Monte Carlo sampling of the fitted parameter covariance matrix. The sampled model variance is combined with the residual variance from the fitted data, then converted to a two-sided interval using a t multiplier when residual degrees of freedom are available.

These bands describe approximate prediction uncertainty around the fitted relationship, not regulatory acceptance limits. They depend on the fitted model form, local covariance approximation, residual behavior, and preprocessing choices.

Prediction uncertainty

Prediction plot error bars: when a prediction dataset is supplied, predicted in vivo times are plotted with horizontal error bars. The bars are calculated from the same 95% prediction-band calculation used for the fitted correlation model; the upper and lower bar lengths are the distances from the band mean to the corresponding upper and lower interval bounds at each input prediction time.

The error bars therefore include propagated fitted-parameter uncertainty and residual/model error from the Approach 1 correlation fit.

12.2 Approach 2: Independent Model Fitting

Fitting uncertainty

Parameter estimates: each in vitro and in vivo model parameter is reported as estimate $\pm$ the approximate 95% confidence-interval half-width. The half-width is calculated from the corresponding nonlinear least-squares covariance matrix as $t(0.975, \text{residual degrees of freedom})$ multiplied by the parameter standard error, with fitted-parameter correlations retained in uncertainty-aware values.

Time constants: tau values, when available, are calculated from the fitted model and parameter covariance. Reported tau values are shown as tau $\pm$ the approximate 95% confidence-interval half-width, calculated as $t(0.975, \text{residual degrees of freedom})$ multiplied by the propagated standard uncertainty.

Fitting plot bands: the in vitro and in vivo shaded 95% prediction bands are generated separately. For each fitted model, the app Monte Carlo samples the fitted parameter covariance matrix, combines sampled model variance with residual variance, and converts the resulting standard uncertainty to a two-sided interval using a t multiplier when residual degrees of freedom are available.

Prediction uncertainty

Value-ratio prediction error bars: uncertainty is propagated from both fitted models. The app samples each model's parameter covariance matrix, calculates the in vivo/in vitro response ratio for each sample, combines ratio variance with residual-based ratio uncertainty, and converts that uncertainty to a two-sided 95% interval. The plotted vertical error bars are scaled from the resulting ratio interval at each prediction time.

Time-constant-ratio prediction error bars: when both tau values are valid, uncertainty in each tau is propagated through the tau ratio using uncertainty-aware arithmetic. The standard uncertainty of the scaled prediction time is multiplied by a t value, where available, and plotted as a symmetric horizontal error bar.

Ratio-based uncertainty can become unstable when the denominator model prediction or denominator time constant is near zero. Predictions in those regions should be interpreted cautiously.

12.3 Approach 3: Direct Mapping

Fitting uncertainty

Parameter uncertainty: Approach 3 does not fit a parametric model, so there is no fitted-parameter covariance matrix, no parameter confidence intervals, and no model-based 95% prediction band.

Fitting plots: the plots show interpolation-based ratios/mappings calculated from the preprocessed datasets. They should be interpreted as deterministic mappings from the supplied data and preprocessing choices, not as statistical uncertainty bands.

Prediction uncertainty

Prediction plot error bars: Approach 3 currently does not generate formal propagated error bars for prediction outputs because no parametric covariance model is fitted.

Uncertainty is therefore limited to the quality of the input data, preprocessing, interpolation, and the assumption that the observed mapping remains valid for the prediction dataset. Extrapolated points outside the interpolation range are excluded or returned as unavailable.

13. Command-line interface

The repository includes `cli.py` for batch-oriented fitting from the command line. Use `python cli.py --help` to see available arguments for input files, approaches, models, metrics, preprocessing options, cross-validation settings, and grid-search settings. The internal keys specified in this user guide are used as settings with `cli.py`.

14. Extending the app

The IVIVC app is designed so that common extensions can be added with minimal changes to the main application code. In many cases, new models, performance metrics, and preprocessing methods can be added by following the structure of the existing files in the relevant subdirectories.

Appendix

A. How to Interpret Model Performance

No single performance metric determines whether an IVIVC model is meaningful. Model selection should consider goodness of fit, prediction error, model complexity, cross-validation behavior, residual patterns, parameter uncertainty, and whether the fitted relationship is scientifically plausible for the polymer degradation mechanism and the intended context of use. For regulatory-facing analyses, the amount and type of supporting evidence should be tied to the model's intended use and credibility needs [1, 2].

The app reports metrics and comparison tables to support user review. It does not select, recommend, validate, or establish regulatory acceptability for any model. Users should document the rationale for the final approach or model selected for a specific analysis.

A.1 Confirm that models are comparable before interpreting metrics

Model-performance metrics are most useful when the compared models were evaluated under the same analysis conditions. In general, compare models only within the same uploaded dataset, preprocessing configuration, response scale, selected IVIVC approach, and candidate-model set.

Do not directly compare AIC, AIC_c, or BIC values across different datasets, transformed responses, preprocessing settings, or substantially different fitting contexts. These information criteria are relative comparison tools, not absolute quality scores. Approach 3 should also be interpreted differently from Approaches 1 and 2 because it uses direct interpolation-based mappings rather than fitting a parametric model.

A.2 Metric ranges, direction, and interpretation

Metric	Range / direction	Interpretation guidance
R^2	Best possible value is 1. Values near 0 indicate little improvement over predicting the mean response. R^2 can be negative when the model performs worse than the mean-response baseline. Higher is better.	R^2 summarizes the fraction of response variance explained by the fitted model [21]. High R^2 does not prove that the model is mechanistic, causal, predictive outside the fitted data range, or appropriate for the intended use.
Adjusted R^2	Higher is better. It can be lower than R^2 and may be negative.	Adjusted R^2 penalizes additional fitted parameters and may be more informative than R^2 when comparing models with different complexity.[11]
MSE	Lower is better. Units are the squared response units.	MSE summarizes average squared error [9, 21]. It is useful computationally but can be less intuitive than RMSE because the units are squared.
RMSE	Lower is better. Units match the response units.	RMSE summarizes typical prediction error on the response scale [9, 21]. Interpret RMSE relative to assay variability, measurement uncertainty, clinically or biologically meaningful differences, and the intended use.
NRMSE	Lower is better. The value is normalized and is intended to be more comparable across response scales.	NRMSE expresses prediction error relative to a normalization quantity, such as the response mean or response standard deviation. Interpretation depends on the normalization method.

Metric	Range / direction	Interpretation guidance
AIC	Lower is better, but only relative to other comparable models. AIC has no fixed lower or upper bound, and negative values are allowed.	AIC balances how well the model fits the data with model complexity [17, 12, 20]. A lower value generally indicates a model representation that explains the data well without unnecessary complexity. Absolute AIC values should not be interpreted as standalone quality grades.
AIC _c	Lower is better, but only relative to other comparable models. AIC _c is undefined when there are too few observations relative to the number of fitted parameters.	AIC _c is AIC with a small-sample correction [22, 12, 20]. It is often preferred when the number of observations is limited relative to the number of parameters, with $n/k < 40$ commonly used as a rule of thumb [12].
BIC	Lower is better, but only relative to other comparable models. BIC has no fixed lower or upper bound, and negative values are allowed.	BIC balances how well the model fits the data with model complexity, usually applying a stronger penalty for additional parameters than AIC [18, 12, 20].
Cross-validation metrics	Direction follows the underlying metric: higher is better for R^2 -type metrics, and lower is better for error metrics.	Cross-validation metrics estimate prediction performance under the selected splitting scheme.[13, 14, 15, 16] They can help identify overfitting, instability, or sensitivity to individual observations or data splits.
Fit/CV ratio	Values near 1 indicate similar fitted-data and cross-validation performance.	For models with consistent fitted and cross-validation performance, the fit/CV ratio is expected to be near 1. Substantial deviations from 1 may indicate reduced generalizability, overfitting, model instability, or sensitivity to the validation procedure, although no universal acceptance threshold is established.

A.3 Interpreting goodness of fit

Goodness-of-fit metrics describe how well the model fits the data used to estimate the model parameters. These metrics include R^2 , adjusted R^2 , MSE, RMSE, and NRMSE.

High fitted-data performance can be encouraging, but it can also be optimistic, especially for small datasets, flexible models, or models with many parameters. A model with strong fitted-data performance may still be unsuitable if it has unstable parameters, poor cross-validation performance, systematic residual patterns, or a fitted shape that is not scientifically plausible for the degradation process.

When interpreting fitted-data metrics, consider whether the uncertainty is small enough to be meaningful for the specific degradation endpoint and context of use. There is no universal performance cutoff for IVIVC adequacy.

A.4 Interpreting prediction error

Prediction-error metrics should be interpreted in practical context. RMSE is often easier to interpret than MSE because it is reported in the same units as the response. NRMSE can be useful for comparing error across response scales, but the denominator matters [10]. For example, NRMSE normalized by the response mean asks, “How large is the typical prediction error compared with the average response value?” whereas NRMSE normalized by the response standard deviation asks, “How large is the typical prediction error compared with the natural spread of the observed responses?” These normalizations answer different questions and should be interpreted carefully when comparing across response scales or datasets.

A prediction error may be numerically small but practically important if the relevant assay, device attribute, or degradation endpoint has narrow acceptable variation. Conversely, a larger numerical error may be acceptable for exploratory model development if the intended use does not require high predictive precision.

A.5 Interpreting relative model evidence

Relative Model Evidence tables summarize information-criterion comparisons among candidate parametric models fitted under comparable conditions. In this context, empirical support means support, within the displayed candidate set, that a model best balances fit quality and complexity. These tables may include AIC, AIC_c, BIC, delta values, evidence weights, and evidence ratios.

Support quantity	Interpretation
AIC, AIC _c , or BIC	Information criterion value. Lower values indicate a better balance of fit quality and model complexity within the displayed candidate set. The absolute value is not interpreted directly.
Delta AIC, Delta AIC _c , or Delta BIC	Difference between a model's criterion value and the lowest value in the displayed candidate set. Values near 0 indicate similar relative support. Larger values indicate lower relative support under that criterion.
Evidence weight	Normalized relative-support value across the displayed candidate models. Weights are comparative summaries, not absolute probabilities that a model is true.
Evidence ratio	Ratio of the highest evidence weight to the model's evidence weight. Larger values indicate greater separation from the highest-weighted model under that criterion.

Common rules of thumb for AIC-family comparisons are: delta values from 0 to 2 suggest similar or substantial empirical support; values from 4 to 7 suggest considerably less support; and values greater than 10 suggest essentially no support relative to the leading candidate in that model set [12]. Common rules of thumb for BIC-family comparisons can also be expressed in support language: delta values from 0 to 2 suggest similar BIC support; values from 2 to 6 suggest lower BIC support; values from 6 to 10 suggest substantially lower BIC support; and values greater than 10 suggest minimal BIC support relative to the leading candidate in that model set [19]. These are interpretive conventions, not regulatory acceptance criteria.

For a given information criterion, the evidence weight for model i can be calculated from the model's delta value:

$$w_i = \frac{\exp\left(-\frac{1}{2}\Delta_i\right)}{\sum_{j=1}^J \exp\left(-\frac{1}{2}\Delta_j\right)} \quad (1)$$

where Δ_i is the difference between model i 's information criterion value and the lowest value among the J candidate models being compared. Larger weights indicate stronger relative support within the candidate model set [12, 20].

Empirical support does not prove that a model is correct, mechanistic, predictive outside the observed range, or suitable for a regulatory submission. If multiple models have similar relative support, users should also consider simplicity, parameter plausibility, residual behavior, cross-validation behavior, and scientific interpretability.

A.6 Interpreting cross-validation performance

Cross-validation estimates prediction performance under the selected splitting pattern. In this app, it is most useful as a sensitivity diagnostic: similar final-fit and cross-validation errors suggest that the fitted relationship is reasonably robust to setting aside parts of the data during repeated fitting, whereas larger differences suggest greater sensitivity to the available observations or to the selected splitting scheme. Cross-validation does not measure parameter uncertainty. The practical question is not only “how good is the score?” but also “what kind of observations did the model have to predict?”

For time-course data, different schemes answer different prediction questions. Leave-one-timepoint-out mainly tests individual omitted points. Leave-contiguous-block-out tests a short missing time window. Ordered K-fold tests larger ordered portions of the dataset. Shuffle split tests randomly omitted observations and is therefore less time-aware. These distinctions matter because adjacent time points may be similar, residual errors may be related across time, and early, middle, and late portions of a degradation profile may represent different behavior. Standard cross-validation procedures are reviewed in [13]; additional considerations for temporally ordered or dependent data are discussed in [14, 15, 16].

For small IVIVC datasets, cross-validation should usually be treated as an exploratory diagnostic or sensitivity analysis. Compare the selected CV metric with the fitted-data metric, but also consider whether the chosen splitting pattern matches the practical prediction question. For example, strong leave-one-timepoint-out performance with weaker contiguous-block performance may suggest that the model can interpolate individual points but is more sensitive when an entire local region is omitted from fitting.

Fit/CV ratios summarize the relationship between fitted-data and cross-validation performance for the same metric. Ratios near 1 indicate that fitted and cross-validation scores are similar. Ratios far from 1 may suggest reduced generalizability, overfitting, model instability, or sensitivity to the splitting scheme. The direction of concern depends on the metric and context, so the ratio should be interpreted together with the raw fit and CV values, the selected splitting scheme, and the fitted plots.

The app also displays cross-validation comparison badges as interpretive aids. These badges are based on the metric's preferred direction: for lower-is-better error metrics (e.g., RMSE), worse cross-validation performance means the CV value is larger than the final-model value; for higher-is-better metrics (e.g., R^2), worse cross-validation performance means the CV value is smaller than the final-model value. As a screening aid, performance within a factor of 1.25 of the final model is labeled as similar, performance worse by 1.25 to 2-fold is labeled as moderately worse, and performance worse by more than 2-fold is labeled as substantially worse. These thresholds are heuristic review aids, not statistical acceptance criteria or regulatory criteria.

The app does not include adjusted R^2 , information criteria (AIC, AIC_c, and BIC), or normalized RMSE (NRMSE) in cross-validation comparison tables. Adjusted R^2 and information criteria depend on the number of observations or fitted parameters, and the final model and cross-validation models are fitted to different subsets of the data. NRMSE is also excluded because its normalization factor may depend on the subset of observations being evaluated, so final-fit and CV values may not use the same denominator. Use adjusted R^2 , information criteria, and NRMSE for fitted-model review and relative empirical support among comparable fitted models, and use eligible cross-validation metrics to evaluate prediction behavior during CV.

A.7 Interpreting residual diagnostics

Residuals (or prediction errors) are calculated as observed minus predicted values. Residual diagnostics are commonly used to assess whether model errors show systematic patterns, nonconstant variance, non-normal behavior, or other departures from model assumptions [23]. The app provides residual-vs-time plots and normal Q-Q plots as qualitative diagnostics.

Residual-vs-time plots keep residuals in the same response units as the fitted endpoint and use a common y-axis scale across comparable models, making it easier to compare residual magnitude visually. Each residual-vs-time plot includes a horizontal zero line and visual reference bands based on a common residual scale. For all comparable models (i.e., models in Approach 1, models in Approach 2), the app calculates the pooled 95th percentile of absolute residuals. The shaded band corresponds to $\pm 25\%$ of that residual scale, and the dashed reference lines correspond to $\pm 50\%$ of that residual scale. These bands provide visual context for comparing residual patterns across models. They are not confidence intervals, prediction intervals, acceptance limits, or regulatory criteria. The common residual scale will depend on the models selected for the Approach.

Normal Q-Q plots compare ordered standardized residuals with the pattern expected under a normal distribution. Points near the diagonal reference line are broadly consistent with normal residual variation, while systematic curvature, large tail departures, or isolated extreme points may suggest non-normality, outliers, or model mismatch. Because IVIVC datasets often contain few time points, Q-Q plots should be interpreted cautiously and used as exploratory diagnostics rather than pass/fail tests.

Residual diagnostics should be reviewed qualitatively. Points scattered around zero in the residual-vs-time plot and near the diagonal line in the Q-Q plot are more consistent with an appropriate model form, although they do not prove that the model is valid. Sustained patterns, curvature, clustering, repeated excursions outside the reference bands, or adjacent time points with residuals mostly on the same side of zero may indicate bias or that the selected model form may not fully capture the data.

A.8 Interpreting parameter uncertainty

Parameter estimates should be reviewed for scientific plausibility and stability. The app reports each available parameter estimate as estimate $\pm$ the approximate 95% confidence-interval half-width, calculated from the parameter

standard error and residual degrees of freedom. Large displayed half-widths, unavailable uncertainty estimates, or highly correlated parameters accompanied by other uncertainty diagnostics may indicate that the data do not strongly identify the model parameters.

The app summarizes fitted-parameter diagnostics using plain-language badges. These badges are screening aids, not acceptance criteria. They are intended to help identify fits that may deserve closer review before a model is used for interpretation or prediction.

Diagnostic	What it summarizes	Interpretation guidance
Relative standard error	Parameter standard error divided by the absolute parameter estimate, expressed as a percentage.	Large relative standard errors can indicate that a parameter is weakly identified or sensitive to the available data. As a screening aid, the app flags values of at least 50% as elevated and at least 100% as large.
Parameter coupling / correlation	Pairwise correlation implied by the parameter-uncertainty calculation.	Correlated parameter estimates can occur naturally in nonlinear models, especially when parameters control scale, shape, or rate. The app reports correlation alone as <i>parameter coupling noted</i> ; when high correlation is accompanied by elevated relative standard error or an internal numerical-sensitivity screening check, the app reports a stronger parameter-identifiability caution. As a screening aid, the app begins noting absolute correlations of at least 0.90, with stronger concern when correlations of at least 0.95 occur together with other uncertainty diagnostics.

If no parameter uncertainty concerns are identified, this means that the app did not detect large relative standard errors or strong parameter coupling using these screening checks. It does not by itself validate the model, prove mechanistic correctness, or establish regulatory acceptability. Conversely, a parameter-coupling or warning badge does not automatically invalidate a model; it indicates that the fitted parameters should be reviewed together with the fitted curves, residuals, cross-validation behavior, model complexity, and scientific plausibility.

Parameter uncertainty is especially important when a fitted model will be interpreted mechanistically, used to support extrapolation, or used to compare degradation behavior across materials, processes, or test conditions. A model with strong fitted metrics but poorly estimated or implausible parameters may be difficult to justify.

A.9 Considering model complexity

More complex models can often fit observed data more closely, but they may overfit small datasets or produce unstable parameter estimates. Adjusted R^2 , AIC, AIC_c, BIC, cross-validation metrics, and residual diagnostics can help evaluate whether additional complexity is justified [24].

When two models have similar fitted performance and similar empirical support, the simpler and more interpretable model may be easier to justify, provided it is scientifically plausible and performs adequately for the intended use. Model complexity should be supported by data density, degradation-mechanism expectations, and the intended context of use.

A.10 Scientific plausibility and intended use

Statistical performance should be interpreted together with knowledge of the polymer, degradation mechanism, test conditions, and proposed use of the IVIVC. A model may be statistically competitive but scientifically difficult to

justify if its fitted shape, parameter values, or extrapolated behavior conflict with known degradation behavior.

The intended use should determine the amount and type of supporting evidence needed. Exploratory model development may tolerate more uncertainty than a model intended to support a specific regulatory argument, design change, or prediction of in vivo performance.

A.11 Suggested model-selection workflow

1. Confirm that the dataset, preprocessing choices, and selected response variables are appropriate.
2. Review fitted curves, uncertainty bands, and residual plots for each candidate model.
3. Compare fitted-data metrics, such as R^2 and RMSE.
4. Review prediction-error metrics in the context of assay variability, endpoint importance, and intended use.
5. Review cross-validation performance and fit/CV ratios, where cross-validation is applicable.
6. Review Relative Model Evidence using AIC, AIC_c, and/or BIC, where parametric model comparisons are applicable.
7. Review fitted-parameter estimates, approximate 95% confidence-interval half-widths, covariance diagnostics, and scientific plausibility.
8. Consider model complexity and whether additional parameters are justified by the data and degradation mechanism.
9. Select the approach or method based on a documented balance of statistical performance, scientific plausibility, and intended use.

A.12 Questions to Guide Model Selection

The following questions may help users interpret model performance in plain language. They are intended as practical prompts for reviewing the results, not as formal acceptance criteria.

A.12.1 Does the model describe the observed data well?

- Does the fitted curve follow the main trend in the observed data?
- Are the fitted-data metrics, such as R^2 , RMSE, and NRMSE, reasonable for the intended use?
- Is the prediction error small enough to be meaningful relative to assay variability, measurement uncertainty, or the degradation endpoint being evaluated?
- Are there obvious regions where the model consistently overpredicts or underpredicts the data?

A.12.2 Does the model appear to generalize beyond the fitted data?

- Are cross-validation metrics similar to the final fitted-model metrics?
- Are fit/CV ratios near 1, or is cross-validation performance substantially worse than fitted-data performance?
- Could the cross-validation result be sensitive to the splitting scheme, number of observations, or a small number of influential points?
- If cross-validation performance is poor, is the model still useful as an exploratory description, or is a simpler or more stable model more appropriate?

A.12.3 Are the model parameters interpretable and stable?

- Are the fitted parameter values scientifically plausible?
- Are the displayed 95% confidence-interval half-widths narrow enough that the parameter estimates appear reasonably precise?
- Are any parameters strongly coupled, meaning that different combinations of parameter values could produce similar fitted curves?

- If parameter coupling is present, does it occur together with large parameter uncertainty or a parameter identifiability caution?
- Is the model being used mainly to describe the curve, or are individual parameter values being interpreted mechanistically?

A model can have a visually reasonable fitted curve while still having uncertain or coupled parameters. In that case, the overall curve may be useful for empirical description within the observed data range, but individual parameter values should be interpreted cautiously.

A.12.4 Is the model unnecessarily complex?

- Does the more complex model improve prediction error, cross-validation behavior, or relative model evidence enough to justify the added parameters?
- Do AIC, AICc, or BIC suggest that the additional complexity is supported within the candidate model set?
- If two models perform similarly, is the simpler model easier to explain, justify, or interpret?
- Does the available dataset contain enough observations to support the number of fitted parameters?

A.12.5 Is the model scientifically plausible for the degradation process?

- Is the fitted curve shape consistent with the expected polymer degradation behavior?
- Are the model assumptions reasonable for the material, test conditions, and degradation endpoint?
- Does the model behave plausibly near the edges of the observed data range?
- Would the model behave plausibly if used for interpolation or extrapolation?

A statistically competitive model may still be difficult to justify if its shape, parameter values, or extrapolated behavior conflict with scientific understanding of the degradation process.

A.12.6 Is the model appropriate for the intended use?

- Is the model being used for exploratory analysis, interpolation, prediction, comparison between conditions, or support for a regulatory rationale?
- How much uncertainty is acceptable for that purpose?
- Are the limitations of the model clearly documented?
- Would an independent reviewer understand why this model or method was selected over reasonable alternatives?

The selected model should be supported by a documented balance of fitted performance, prediction error, cross-validation behavior, relative empirical support, parameter diagnostics, residual behavior, scientific plausibility, and intended use.

A.13 Example interpretation statements

The following examples illustrate wording that may be adapted for reports. Users should tailor the language to the specific analysis and supporting evidence.

- Model A had the lowest RMSE and highest adjusted R^2 among the evaluated models, indicating strong performance under the selected preprocessing conditions.
- Model B had similar AIC_c support to Model A, with Delta AIC_c less than 2, suggesting that the two models were not clearly distinguished by AIC_c within this candidate set.
- Although Model C had strong fitted-data performance, its cross-validation error was substantially higher than its fitted error, suggesting possible sensitivity to the validation procedure or reduced generalizability.
- The selected model was chosen based on a balance of fitted performance, cross-validation behavior, relative empirical support, parameter plausibility, and scientific interpretability.

A.14 Limitations

Performance metrics are descriptive and comparative tools, not acceptance criteria. Small datasets can produce unstable estimates, and similar metric values do not necessarily indicate equivalent scientific validity. Users should document preprocessing assumptions, fitting settings, model selection rationale, limitations, and any planned use of the model outside the fitted data range.

References

- [1] U.S. Food and Drug Administration. Assessing the credibility of computational modeling and simulation in medical device submissions: Guidance for industry and food and drug administration staff. <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/assessing-credibility-computational-modeling-and-simulation-medical-device-submissions>, November 2023.
- [2] U.S. Food and Drug Administration. Reporting of computational modeling studies in medical device submissions: Guidance for industry and food and drug administration staff. <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/reporting-computational-modeling-studies-medical-device-submissions>, September 2016.
- [3] Harry B. Hopfenberg. Controlled release from erodible slabs, cylinders, and spheres. In *Controlled Release Polymeric Formulations*, volume 33 of *ACS Symposium Series*, pages 26–32. American Chemical Society, 1976. <https://doi.org/10.1021/bk-1976-0033.ch003>.
- [4] A. Tsoularis and J. Wallace. Analysis of logistic growth models. *Mathematical Biosciences*, 179(1):21–55, 2002. [https://doi.org/10.1016/S0025-5564\(02\)00096-2](https://doi.org/10.1016/S0025-5564(02)00096-2).
- [5] Achim Göpferich. Polymer bulk erosion. *Macromolecules*, 30(9):2598–2604, 1997. <https://doi.org/10.1021/ma961627y>.
- [6] Susanne Fredenberg, Marie Wahlgren, Mats Reslow, and Anders Axelsson. The mechanisms of drug release in poly(lactic-co-glycolic acid)-based drug delivery systems—a review. *International Journal of Pharmaceutics*, 415(1–2):34–52, 2011. <https://doi.org/10.1016/j.ijpharm.2011.05.049>.
- [7] S. Dean Allison. Analysis of initial burst in PLGA microparticles. *Expert Opinion on Drug Delivery*, 5(6):615–628, 2008. <https://doi.org/10.1517/17425247.5.6.615>.
- [8] M. H. Zwietering, I. Jongenburger, F. M. Rombouts, and K. van ’t Riet. Modeling of the bacterial growth curve. *Applied and Environmental Microbiology*, 56(6):1875–1881, 1990. <https://doi.org/10.1128/aem.56.6.1875-1881.1990>.
- [9] Timothy O. Hodson. Root-mean-square error (RMSE) or mean absolute error (MAE): When to use them or not. *Geoscientific Model Development*, 15:5481–5487, 2022. <https://doi.org/10.5194/gmd-15-5481-2022>.
- [10] Rob J. Hyndman and Anne B. Koehler. Another look at measures of forecast accuracy. *International Journal of Forecasting*, 22(4):679–688, 2006. <https://doi.org/10.1016/j.ijforecast.2006.03.001>.
- [11] Jeremy Miles. R squared, adjusted r squared. In *Wiley StatsRef: Statistics Reference Online*. John Wiley & Sons, 2014. <https://doi.org/10.1002/9781118445112.stat06627>.
- [12] Kenneth P. Burnham and David R. Anderson. *Model Selection and Multimodel Inference: A Practical Information-Theoretic Approach*. Springer, 2nd edition, 2002. <https://doi.org/10.1007/b97636>.
- [13] Sylvain Arlot and Alain Celisse. A survey of cross-validation procedures for model selection. *Statistics Surveys*, 4:40–79, 2010. <https://doi.org/10.1214/09-SS054>.
- [14] Christoph Bergmeir and José M. Benítez. On the use of cross-validation for time series predictor evaluation. *Information Sciences*, 191:192–213, 2012. <https://doi.org/10.1016/j.ins.2011.12.028>.
- [15] David R. Roberts, Volker Bahn, Simone Ciuti, Mark S. Boyce, Jane Elith, Gurutzeta Guillera-Arroita, Severin Hauenstein, José J. Lahoz-Monfort, Boris Schröder, Wilfried Thuiller, David I. Warton, Brendan A. Wintle, Florian Hartig, and Carsten F. Dormann. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8):913–929, 2017. <https://doi.org/10.1111/ecog.02881>.
- [16] Christoph Bergmeir, Rob J. Hyndman, and Bonsoo Koo. A note on the validity of cross-validation for evaluating autoregressive time series prediction. *Computational Statistics & Data Analysis*, 120:70–83, 2018. <https://doi.org/10.1016/j.csda.2017.11.003>.
- [17] Hirotugu Akaike. A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6):716–723, 1974. <https://doi.org/10.1109/TAC.1974.1100705>.
- [18] Gideon Schwarz. Estimating the dimension of a model. *The Annals of Statistics*, 6(2):461–464, 1978. <https://doi.org/10.1214/aos/1176344136>.

- [19] Robert E. Kass and Adrian E. Raftery. Bayes factors. *Journal of the American Statistical Association*, 90(430):773–795, 1995. <https://doi.org/10.1080/01621459.1995.10476572>.
- [20] Eric-Jan Wagenmakers and Simon Farrell. AIC model selection using akaike weights. *Psychonomic Bulletin & Review*, 11:192–196, 2004. <https://doi.org/10.3758/BF03206482>.
- [21] scikit-learn developers. Regression metrics documentation. https://sklearn.org/stable/modules/model_evaluation.html#regression-metrics.
- [22] Clifford M. Hurvich and Chih-Ling Tsai. Regression and time series model selection in small samples. *Biometrika*, 76(2):297–307, 1989. <https://doi.org/10.1093/biomet/76.2.297>.
- [23] Michael H. Kutner, Christopher J. Nachtsheim, John Neter, and William Li. *Applied Linear Statistical Models*. McGraw-Hill/Irwin, 5 edition, 2005.
- [24] Georg Heinze, Christine Wallisch, and Daniela Dunkler. Variable selection – a review and recommendations for the practicing statistician. *Biometrical Journal*, 60(3):431–449, 2018. <https://doi.org/10.1002/bimj.201700067>.